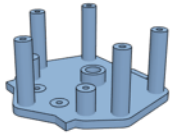
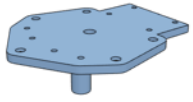
# Parts

Print these (STEP in ../STEP). Right-hand build.



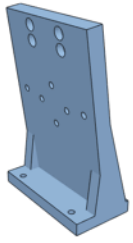
base_bottom

**1×**



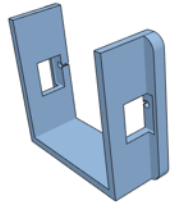
base_top

**1×**



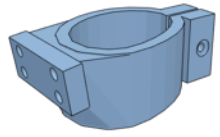
tab_mast

**1×**



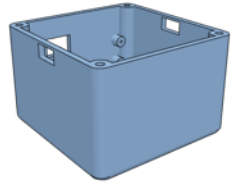
d405_hood

**1×**



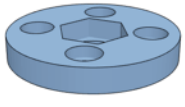
quest_arm_right

**1×**



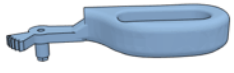
ebox_cover

**1×**



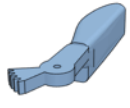
servo_coupler

**1×**



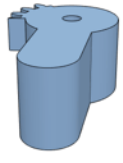
handle_a_rh

**1×**



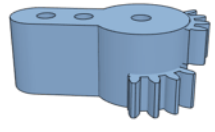
handle_b_rh

**1×**



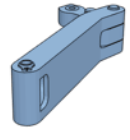
knuckle_gear_left

**1×**



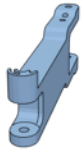
knuckle_gear_right

**1×**



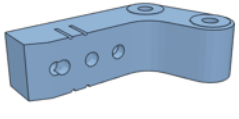
outer_link_2f140

**2×**



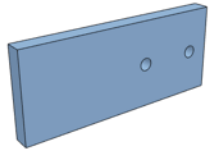
inner_knuckle_2f140

**2×**



inner_finger_2f140

**2×**



pad_core_2f140

**2×**

pad_skin_2f140

**2×**

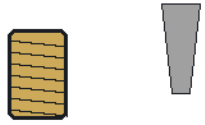
grip_tpu_a_rh

**1×**

grip_tpu_b_rh

**1×**

# 0



Press each insert in with a soldering iron at ~220 °C, flush with the surface, before assembling.

## Heat-set inserts

| Part | Inserts | Where |
|---|---|---|
| base_bottom | **9 × M3** | 5 standoff tops, 4 underneath for the box |
| base_top | **4 × M3** | the mast's foot |
| handle_a_* | **1 × M4 + 1 × M3** | axle bolt (top of the hub), stop screw (underneath) |
| knuckle_gear_left / _right | **2 × M3 each** | the key screws |
| pad_core_2f140 | **2 × M3 each** | the pad screws |
| quest_arm_* | **4 × M3** | the pad on the mast's back |

Total: 26 × M3 (M3 × 5.7) and 1 × M4. Glue the TPU grip inserts into their handles (CA glue).

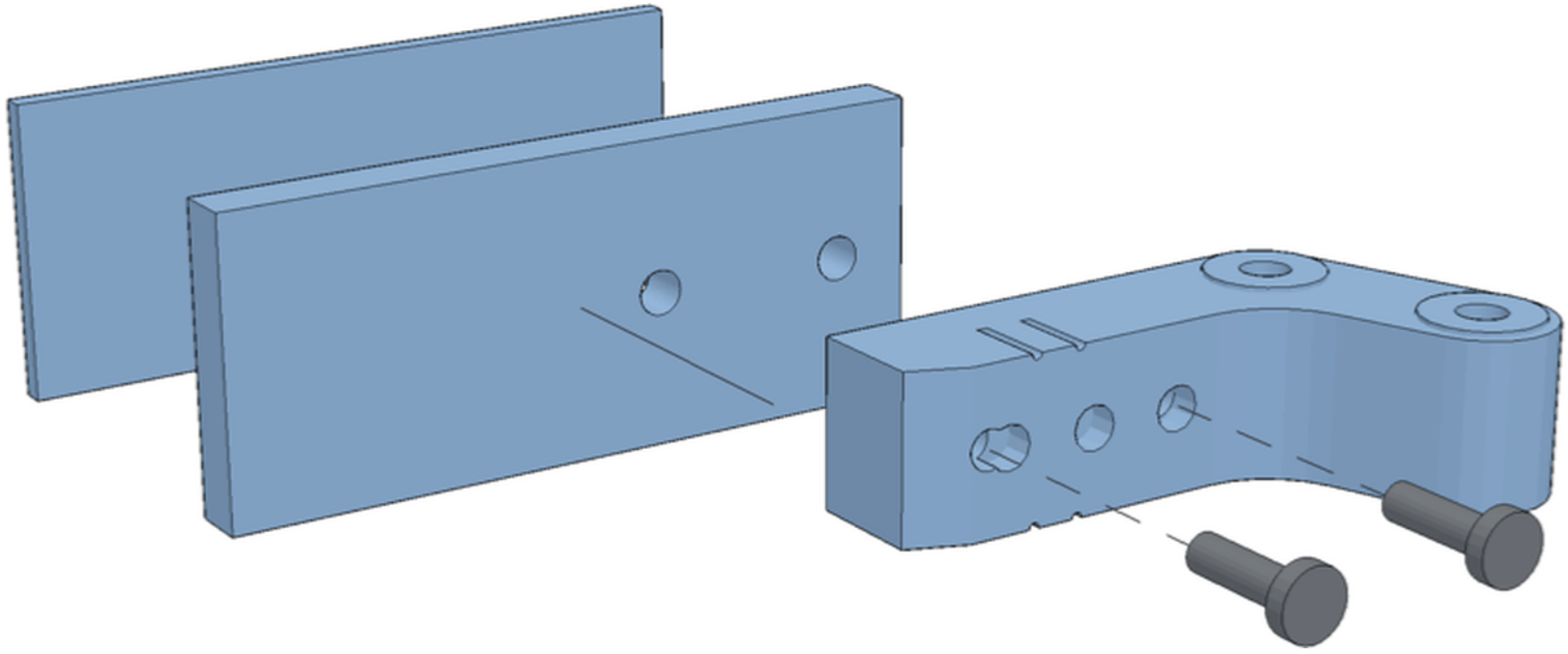
# 1

## Pads

2×

1× TPU skin: CA glue

2× M3 × 10

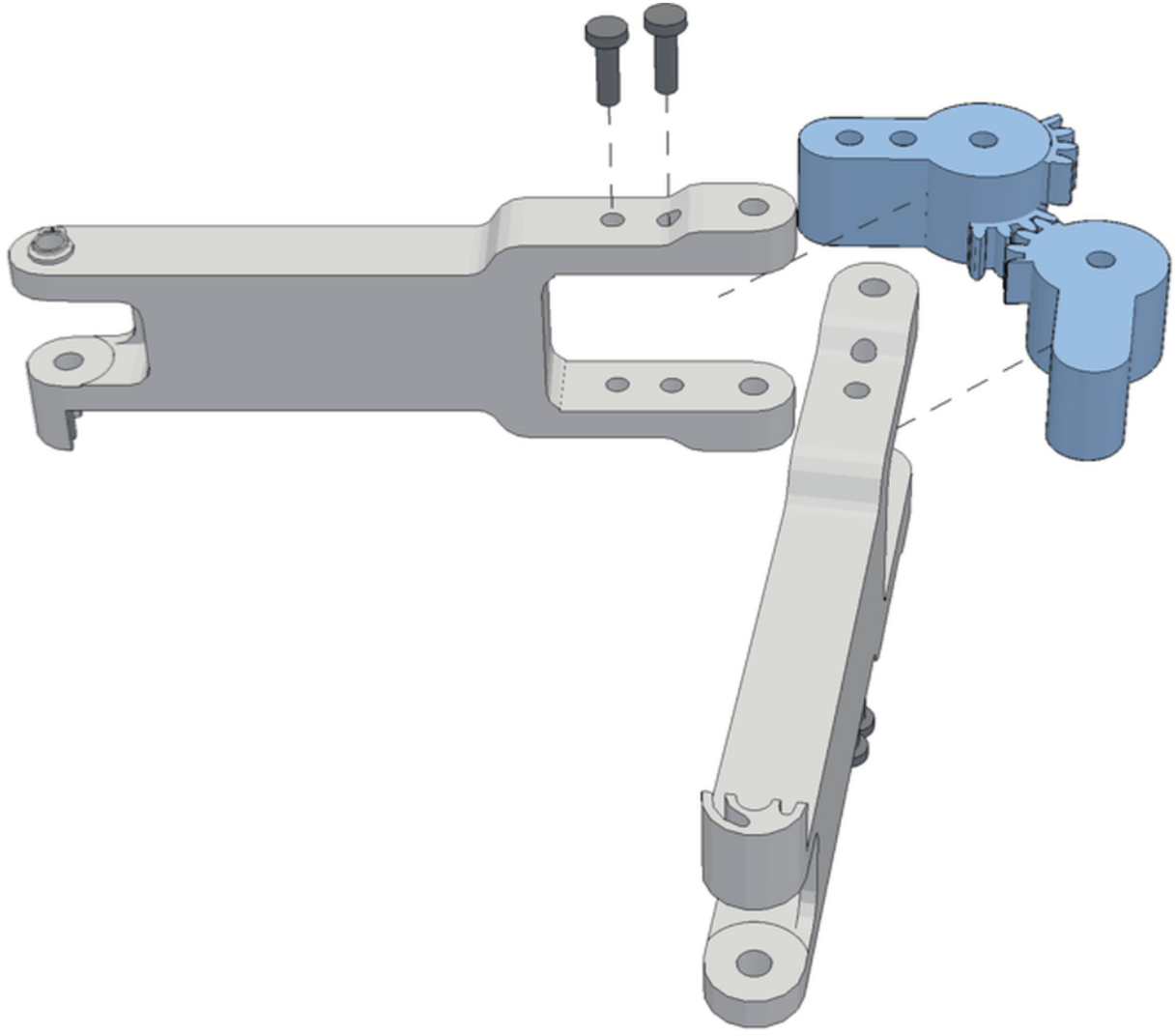


Glue the TPU skin to the core. Screw the core to the inner finger. Make two.

# 2

## Knuckle gears

4×
M3 × 10



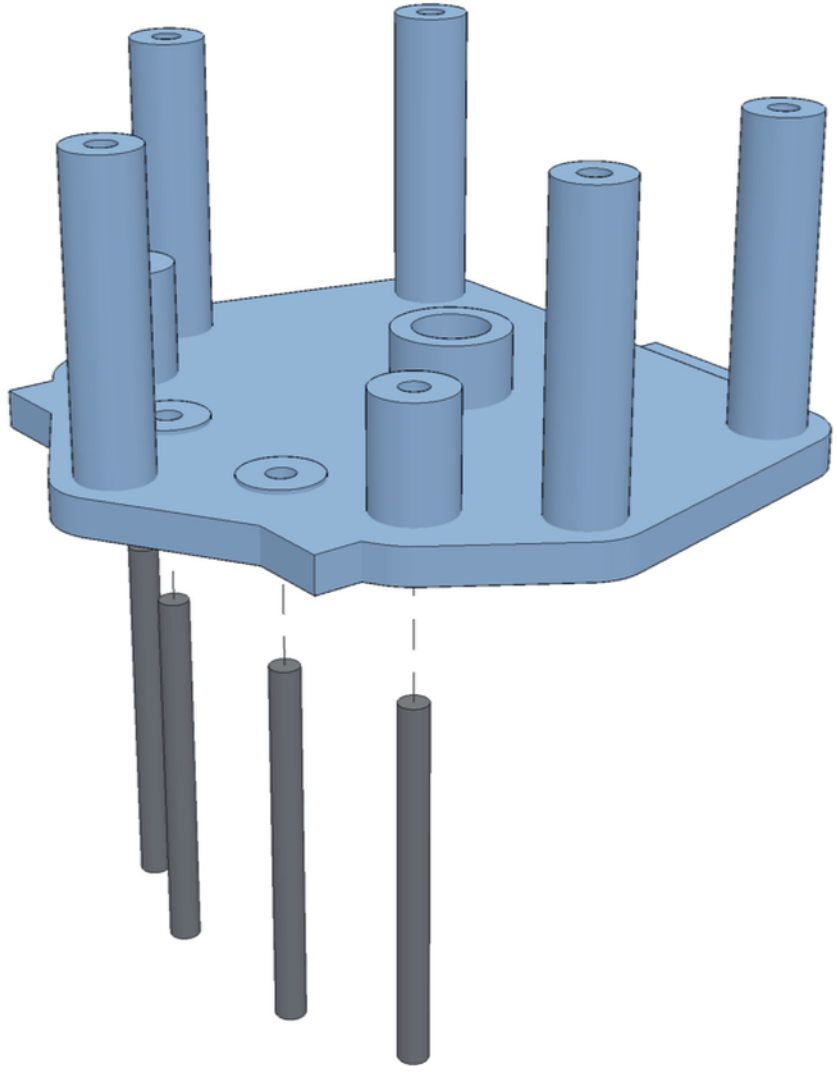
Slide each gear into its inner knuckle's fork. Left gear: screws from below. Right gear: from above.

# 3

## Bottom plate

**4×** Ø4 × 50 dowel

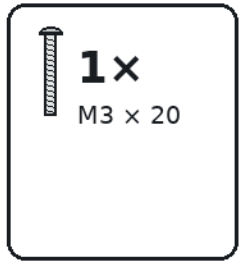
**1×** 12 × 12 tactile switch

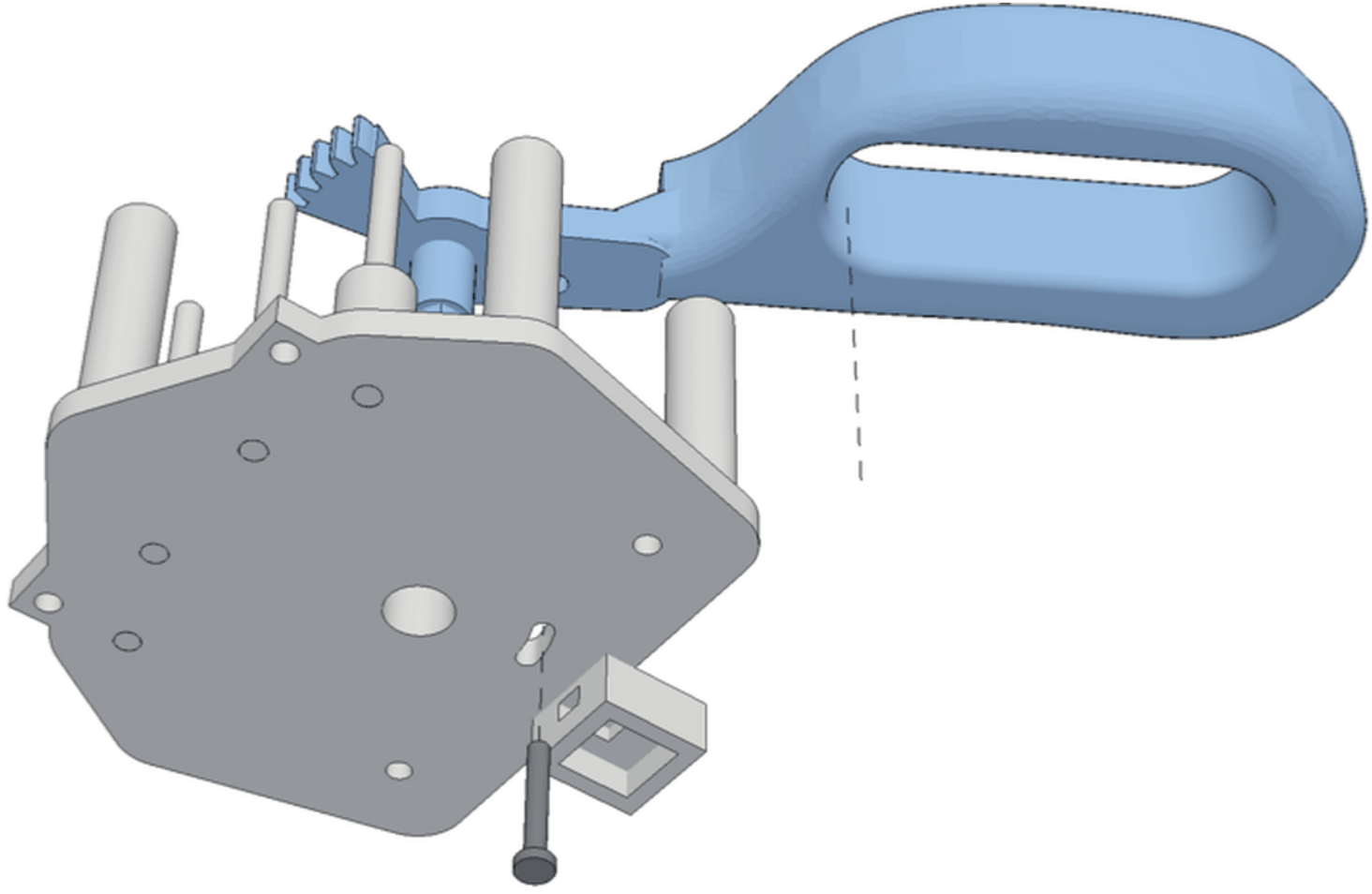


Press the four long dowels into the plate (a vice helps). Push the switch into the pod underneath, button down.

# 4

## Handle A

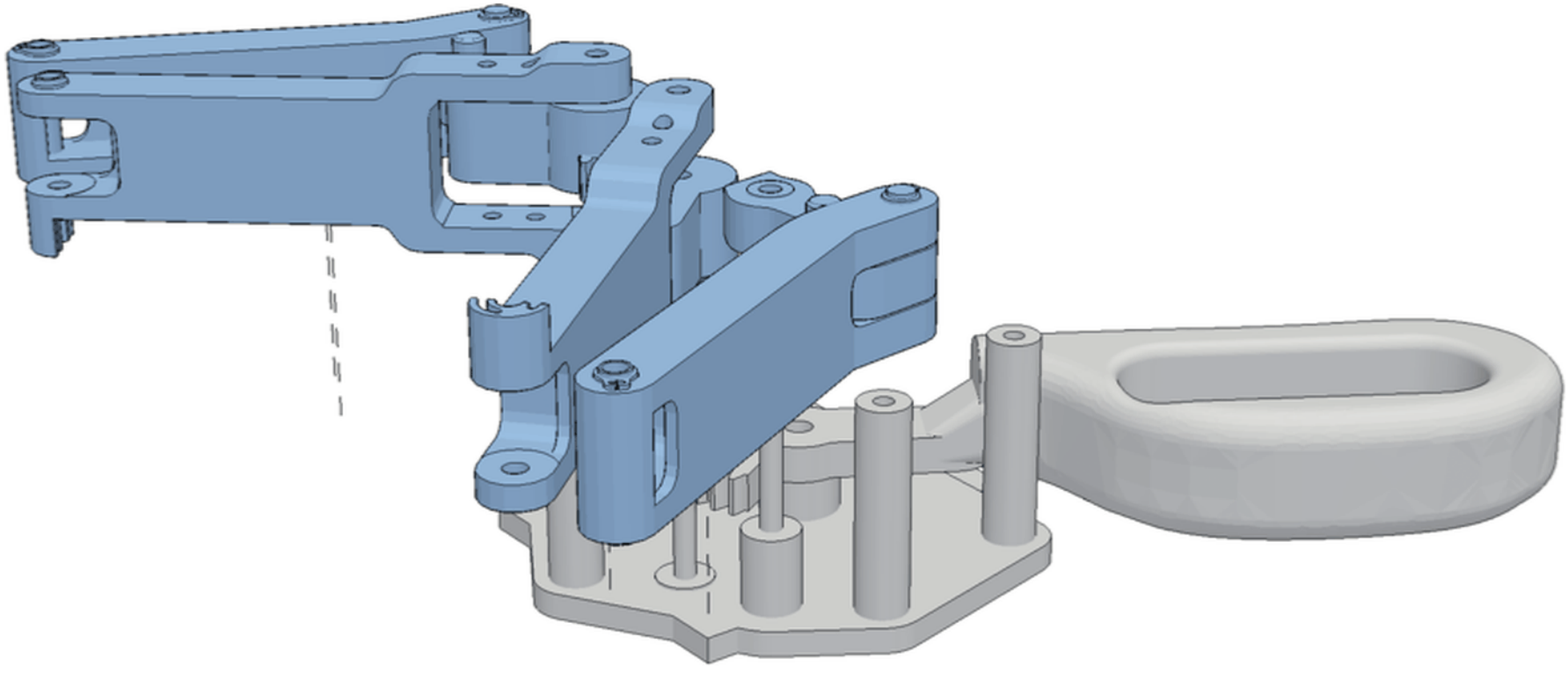


Drop handle A's hub into the bottom plate's bearing. From below, screw the stop screw through the arc slot into handle A.

# 5

## Jaws

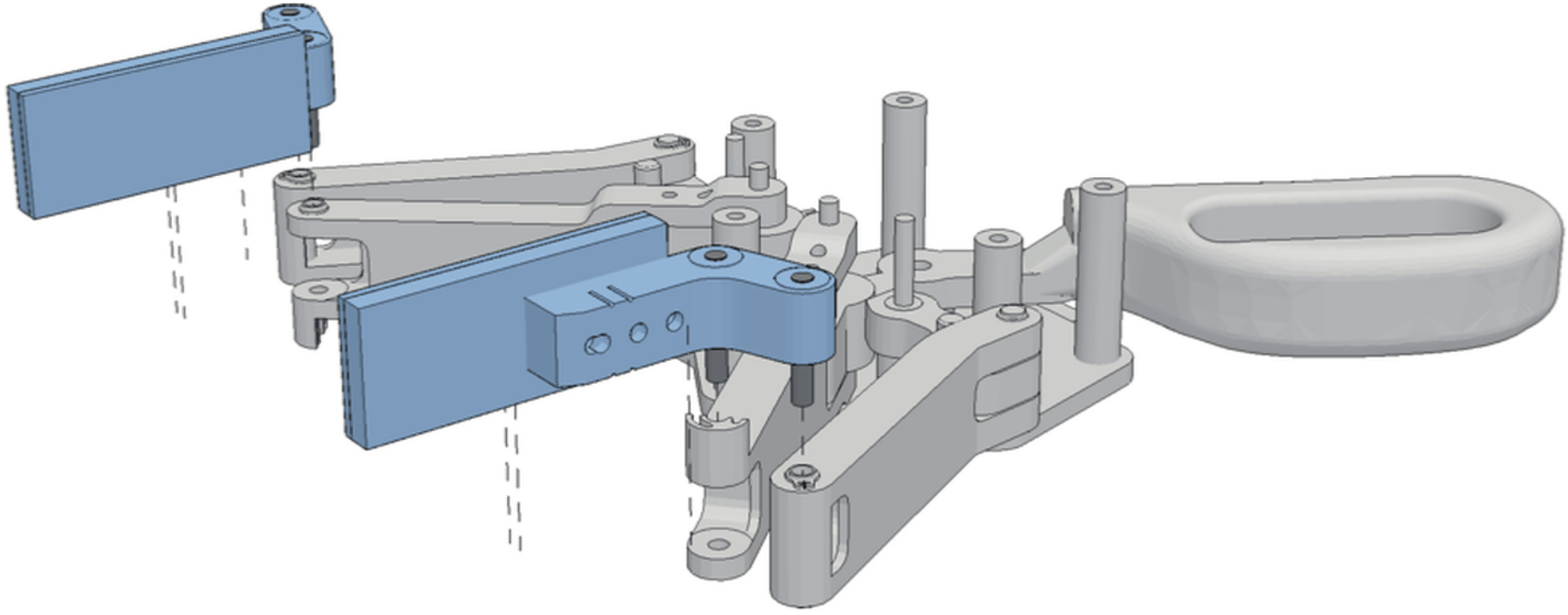


Fully open: hold handle A against its open stop. Slide both outer links and both knuckle + gear assemblies onto the dowels so the gears mesh with each other and with handle A.

# 6

## Fingers

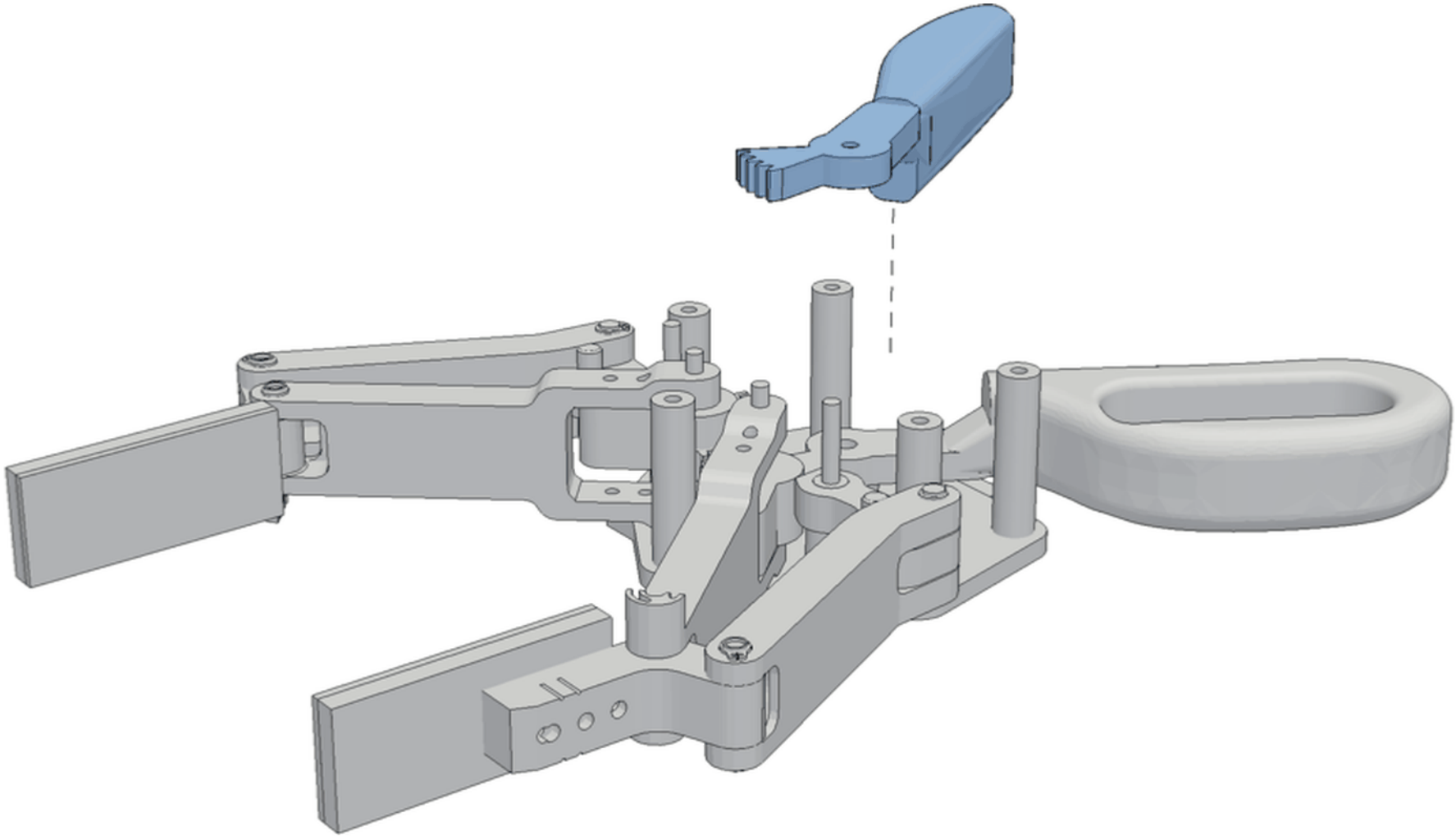
**4×**
Ø4 × 25 dowel



Fit each finger (with its pad) to its outer link and inner knuckle and pin both joints.

# 7

## Handle B



With the jaws still open, lay handle B on the axle so its teeth mesh with the right gear.

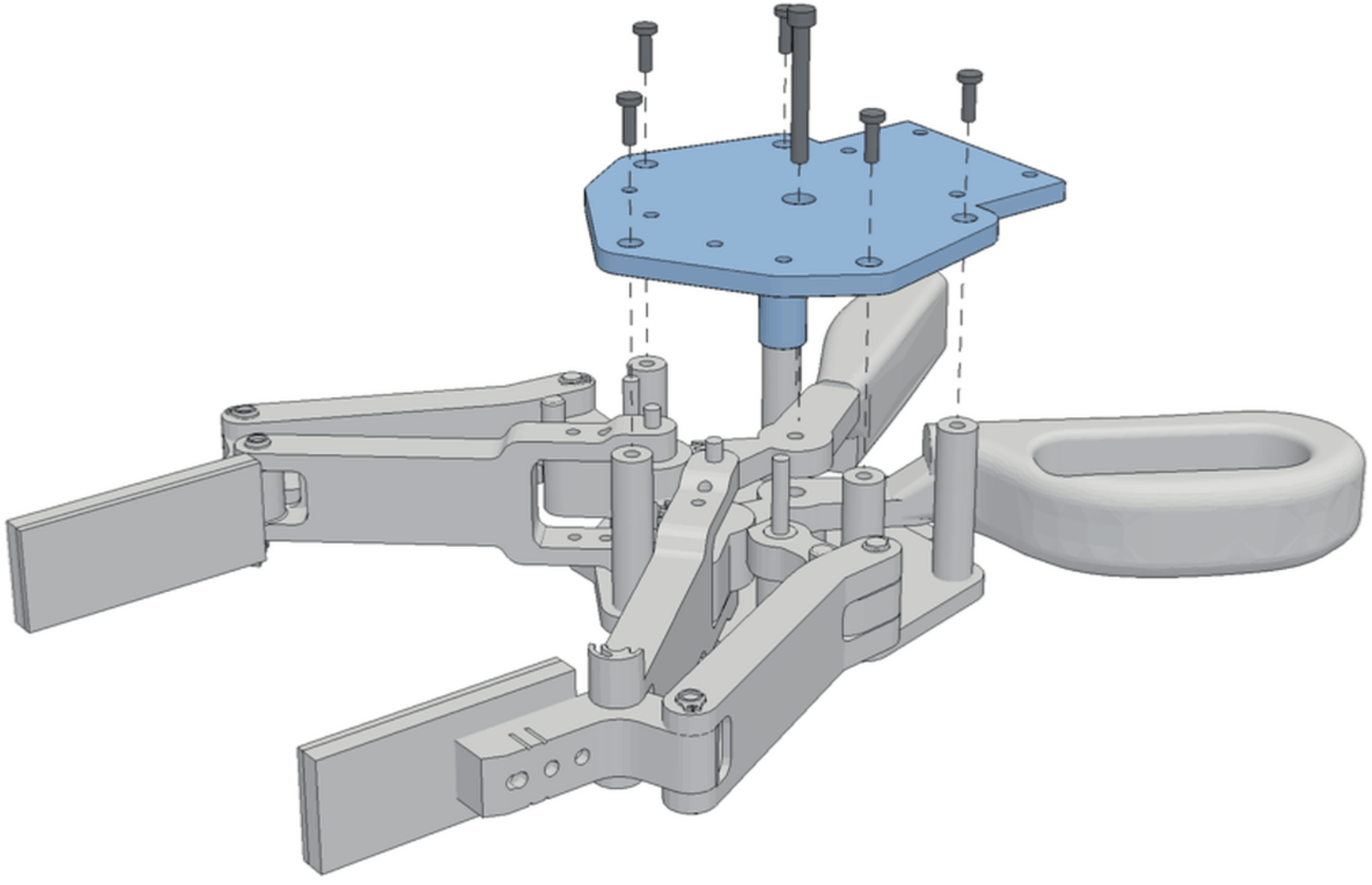
# 8

## Top plate

**5×** M3 × 10

**1×** M4 × 35

**1×** threadlocker on the M4

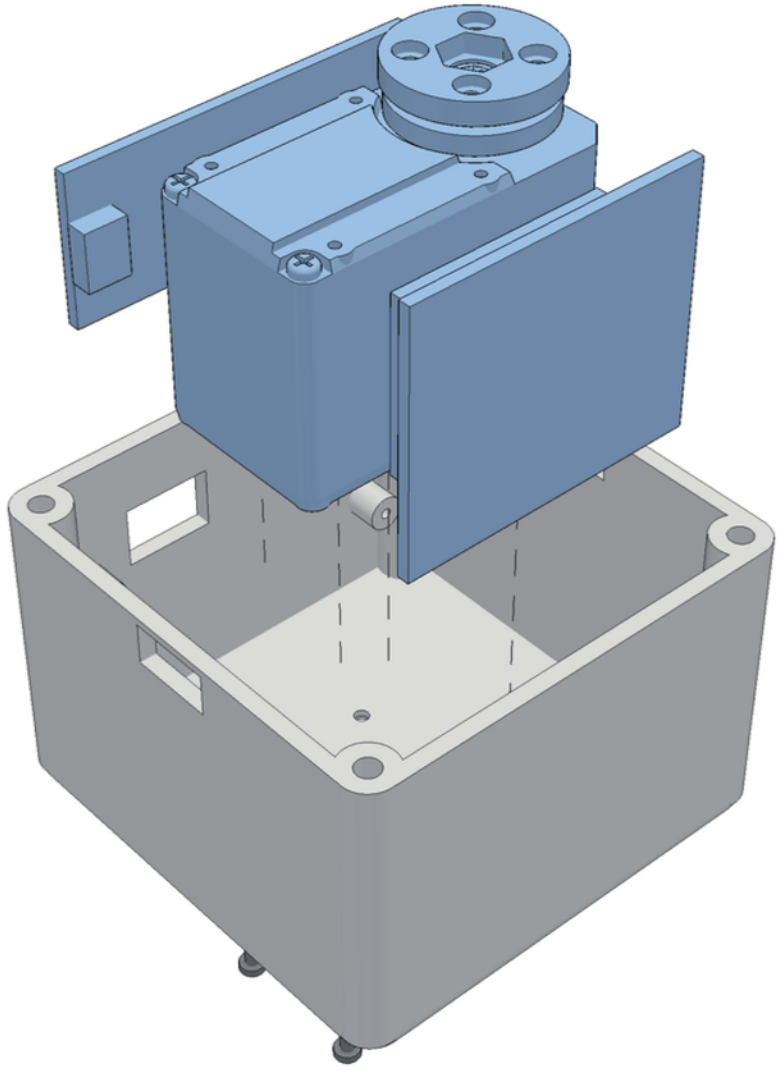


Fit the top plate over the dowels. Screw it to the five standoffs. Axle bolt through the top plate and handle B into handle A, with threadlocker; snug, so the handles still turn freely.

# 9

## Electronics

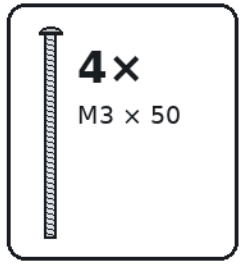
| | | |
|---|---|---|
| **4×** | **4×** | **4×** |
| M2 × 8 | M2 × 6 | M2 × 5 self-tapping |

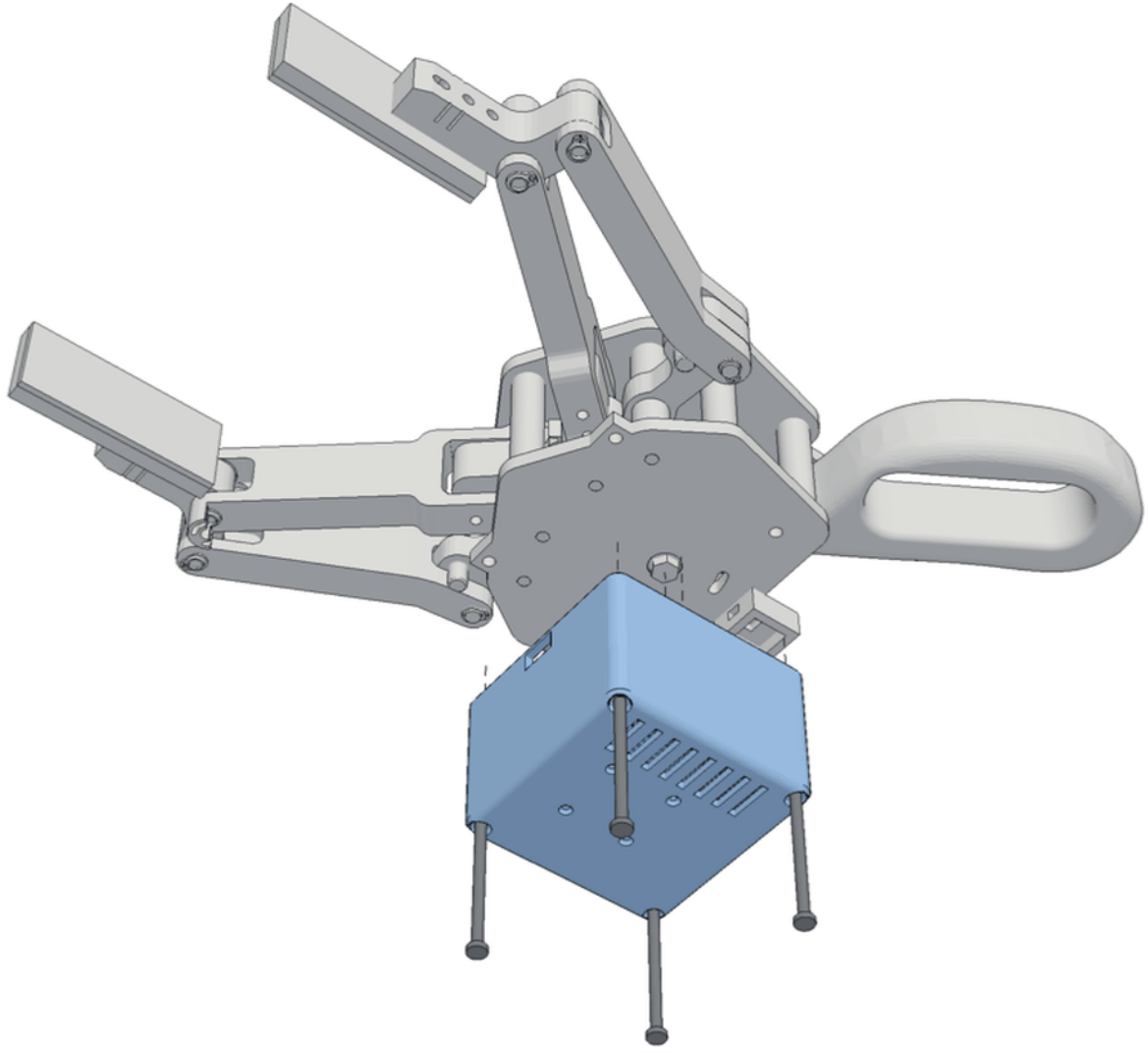


Servo on the box floor (4 × M2 × 8 from below). Coupler on the servo horn (4 × M2 × 6). Pico 2 on its standoffs, hub in its bay, IMU on the back wall. Wire them as in electronics.md.

# 10

## Electronics box



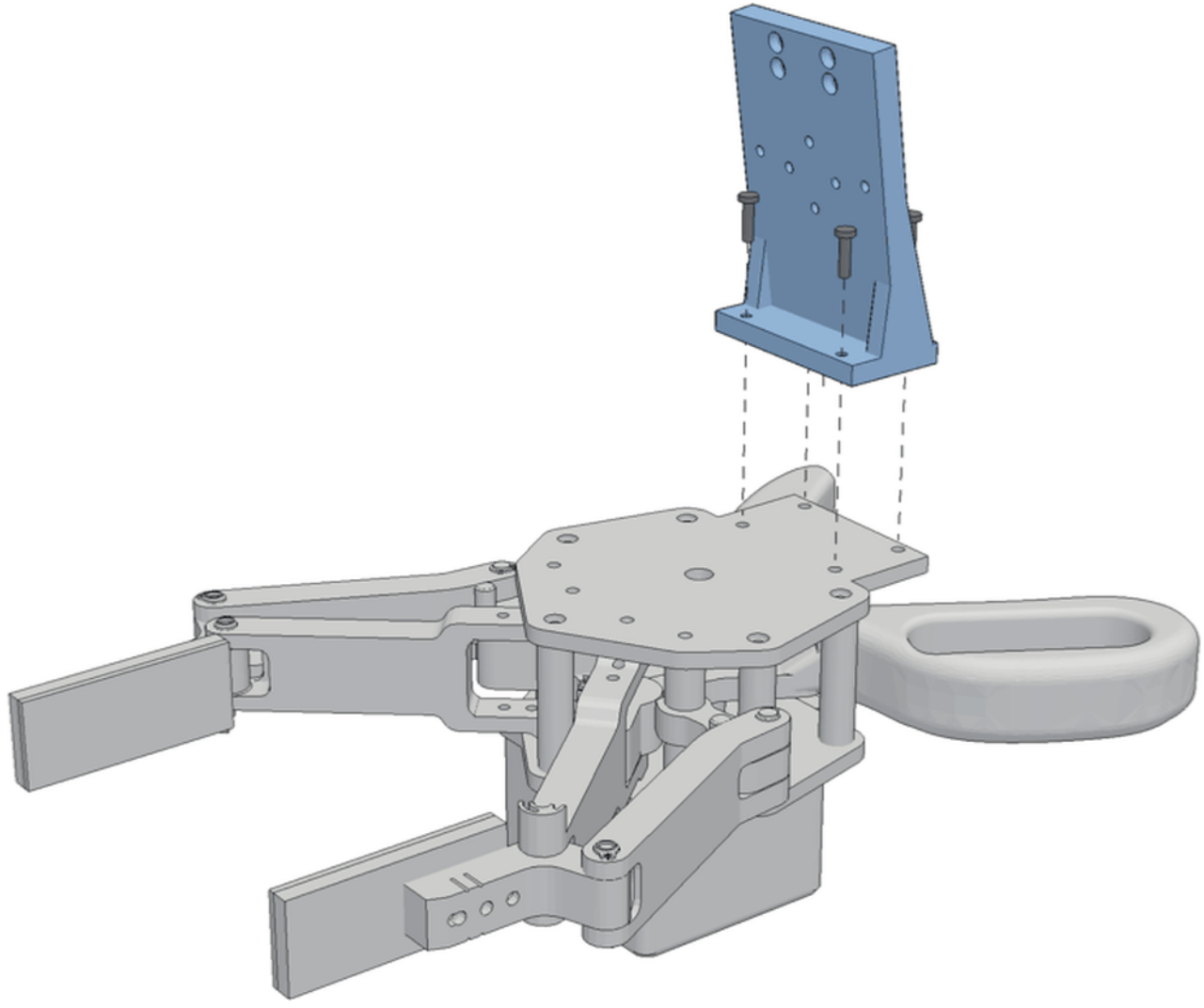



Jaws open, servo at its open position: push the box up so handle A's hex enters the coupler. Screw the box to the bottom plate.

# 11

## Camera mast

**4×**
M3 × 12

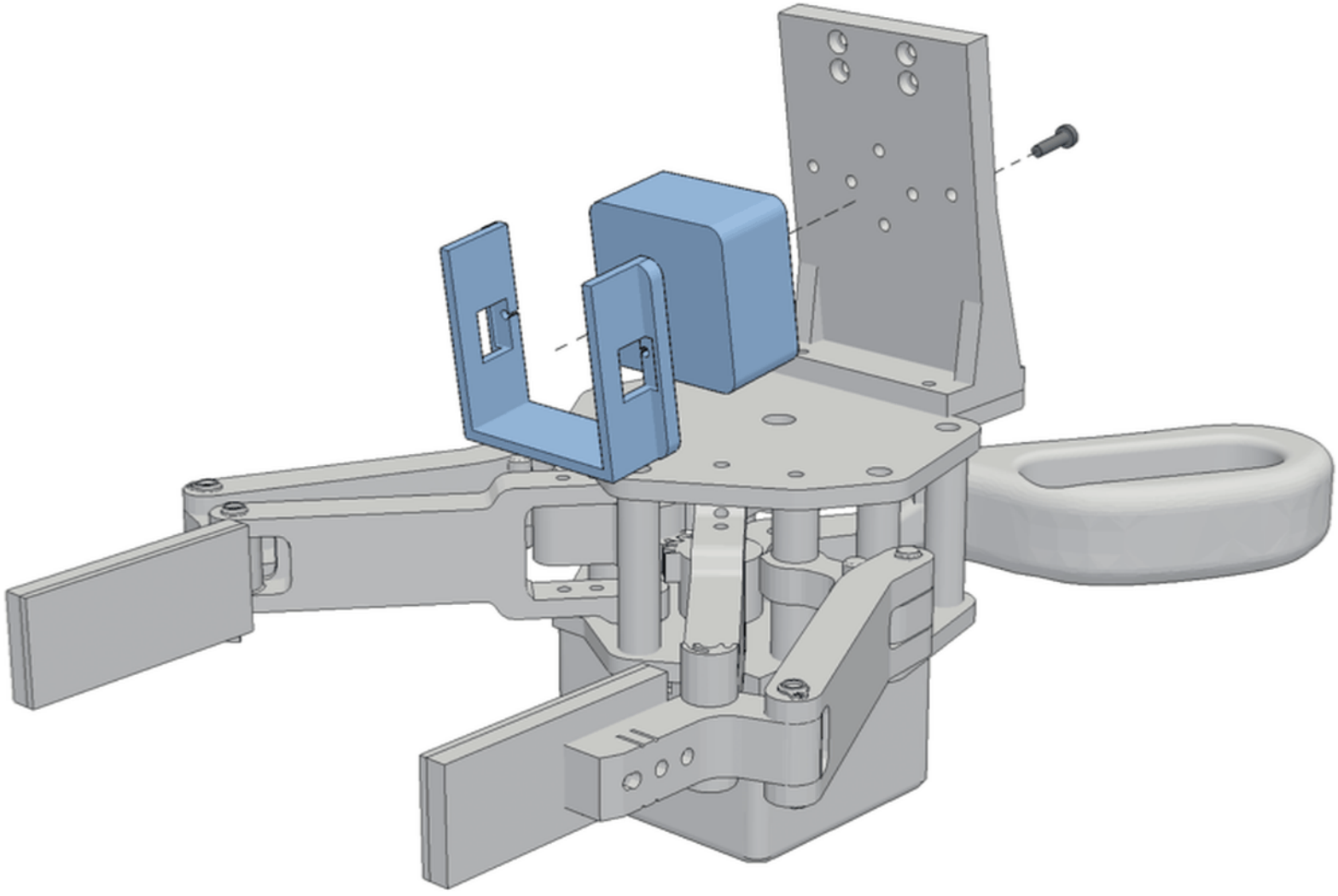


Screw the mast's foot to the top plate.

# 12

## D405

**2×**
M3 × 12 self-tapping



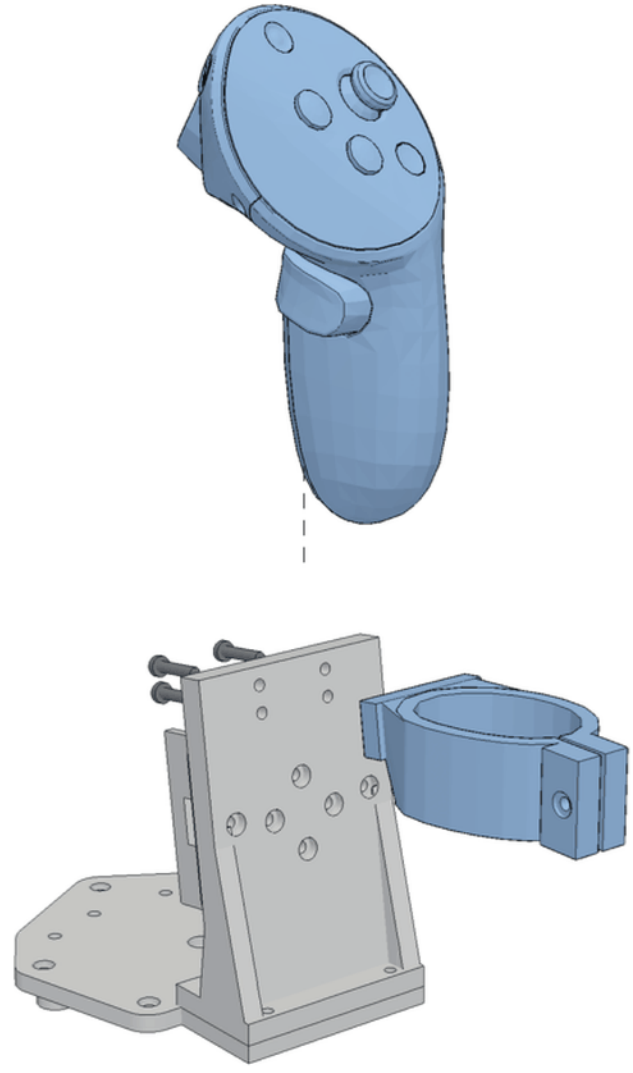
Fix the D405 to the tab exactly as on the robot's camera plate. Put the hood over it and screw it on from behind the mast.

# 13

## Quest controller

**4×** M3 × 16

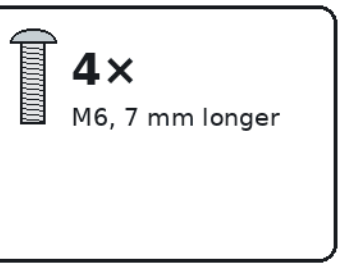
**1×** M3 × 20 + nut



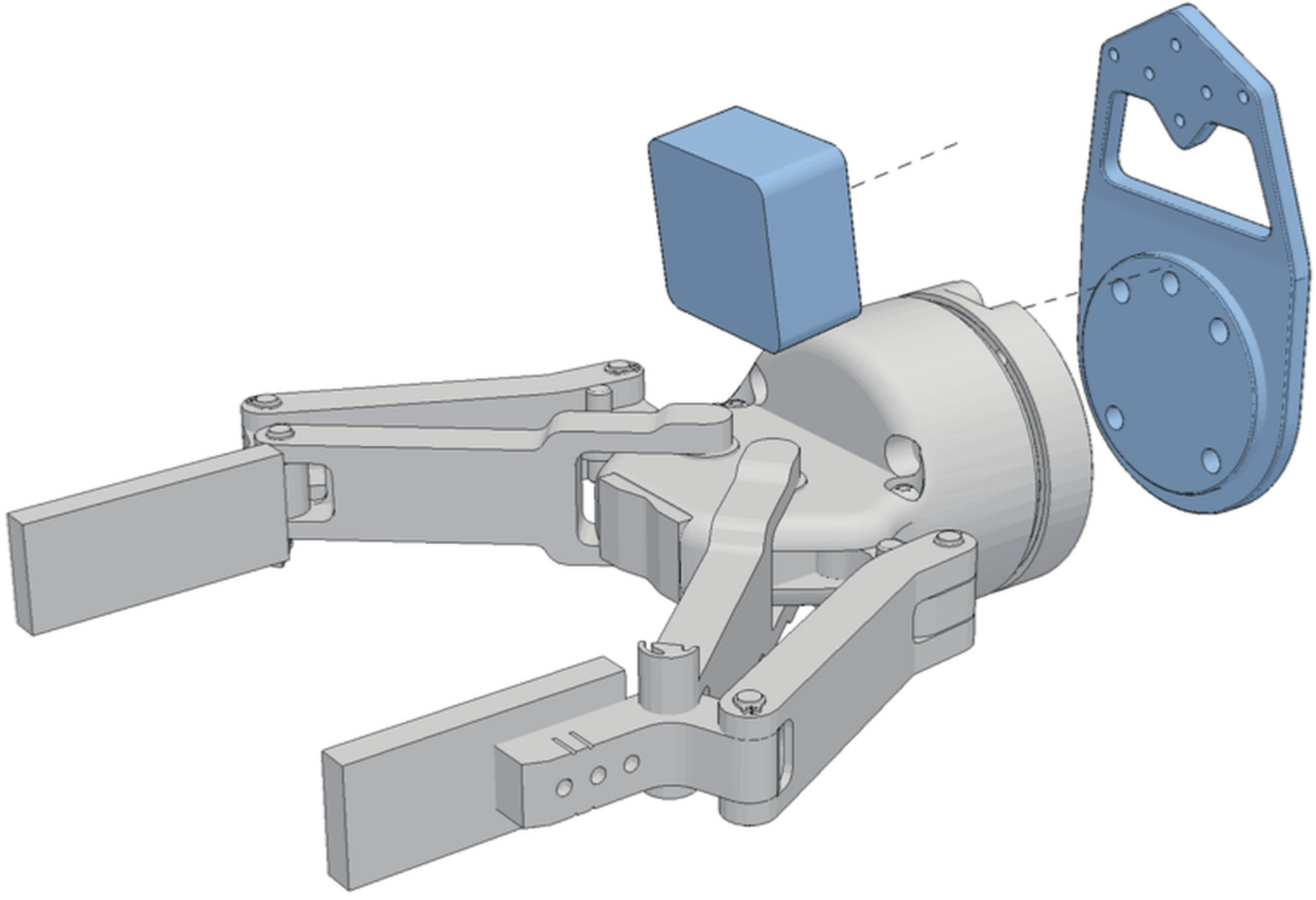
Screw the right Quest arm to the mast's back from the front. Push the right Touch Pro's handle into the collar, trigger forward, and clamp it.

# 14

## Robot



**4×**
M6, 7 mm longer



On the robot: the Robotiq 2F-140 as it is. Put our camera plate between its coupling and the UR flange (M6 screws 7 mm longer than the coupling's own) and fix the D405 to the plate's tab.